

ROBERT ANADU

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OBJECTIVE

Aerospace Engineering graduate specializing in **flight dynamics, control systems, and avionics integration**. Experienced in designing and validating **simulation-driven aerospace systems** using **MATLAB/Simulink and Python**. Seeking an entry-level role in **aerospace systems, flight controls, or avionics engineering**.

EDUCATION

Toronto Metropolitan University (formerly Ryerson University)

Toronto, ON

Bachelor of Engineering in Aerospace Engineering

Sept 2020 – Jun 2024

Relevant Coursework: Control Systems, Flight Dynamics, Avionics Design, Space Robotics, Machine Learning, Structural Analysis, Fluid Mechanics

CERTIFICATIONS

Google Cybersecurity Professional Certificate

2025

Coursera

Online

Completed hands-on labs covering network security fundamentals, threat modeling, and secure system architecture concepts applicable to safety-critical systems.

TECHNICAL SKILLS

Core Engineering Tools: MATLAB, Simulink, SolidWorks, CATIA, ANSYS, AutoCAD

Programming & Software: Python, MATLAB, C/C++, JavaScript, TypeScript, SQL, LaTeX

Aerospace & Controls: Flight Dynamics, Flight Control Systems, Avionics Integration, State-Space Modeling, PID Design & Tuning, Aircraft Performance

Systems & Technologies: FastAPI, React, OpenCV, YOLOv8, Embedded Systems (Arduino), Git

EXPERIENCE

Facilities Technical Support Technician

Dec 2024 – Present

BGIS

Toronto, ON

- Interpreted **engineering drawings and specifications** to execute precision installations of modular mechanical systems under strict spatial and safety constraints.
- Performed **on-site system assessments**, layout verification, and fault identification to ensure operational and safety compliance.
- Applied **structured problem-solving and process planning** to support facility system upgrades and continuous improvement initiatives.
- Operated in **time-critical environments** requiring coordination across multiple trades while maintaining quality standards.

PROJECTS

Capstone Project — Autonomous Vision-Guided Helicopter with Remote Payload

Apr 2024

Toronto Metropolitan University

Toronto, ON

- Designed and simulated a **multi-layer avionics architecture** for a **3-DOF helicopter** integrating guidance, navigation, control, vision processing, and payload operation.
- Developed **quintic (5th-order) trajectory planning algorithms** with velocity-based timing to achieve smooth, dynamically feasible autonomous motion within **actuator and stability constraints**.
- Integrated a **camera-based vision system** for target localization and conversion of pixel measurements into navigation-relevant angular commands.

Supersonic Propulsion System Design & Trade Study

Mar 2024

Toronto Metropolitan University

Toronto, ON

- Designed a **Mach 3.2 propulsion system** integrating an **Oswatitsch-optimized multi-shock supersonic inlet** with a one-dimensional turbojet cycle model.
- Conducted **parametric trade studies** across compressor pressure ratio and turbine inlet temperature to balance specific thrust, TSFC, and ram-drag at supersonic cruise.
- Benchmarked performance against the **Rolls-Royce/Snecma Olympus 593**, highlighting limitations of Mach 2-optimized engines at Mach 3.2.

Aircraft Structural Redesign: Wing Box & Landing Gear

Apr 2023

Toronto Metropolitan University

Toronto, ON

- Designed and validated **conceptual wing box and landing gear structures** using parametric CAD, section analysis, and first-order load calculations to establish realistic load paths.
- Developed **manufacturable structural layouts** emphasizing internal geometry, wall thickness selection, and structural clarity appropriate for early-stage aerospace design.
- Produced **professional engineering documentation and visualizations**, including orthographic views and section cuts, to communicate structural intent and design assumptions.

AI Chatbot Assistant (Full-Stack Project)

Jan 2026

Personal Project

Toronto, ON

- Built a **full-stack AI chatbot** using **FastAPI (Python)** and **React + TypeScript**, enabling real-time conversational interaction.
- Implemented **persistent conversational memory** using SQLite to support multi-turn contextual responses and state management.
- Designed a **modern, Apple-style user interface** and deployed the system using **Vercel** (frontend) and **Render** (backend).

LEADERSHIP AND VOLUNTEER EXPERIENCE

Sound Engineer Volunteer

Sept 2015 – Mar 2020

Various Institutions

Lagos, Nigeria – Toronto, ON

Led live audio operations for events with 100+ attendees and mentored junior technicians.

Demonstrated teamwork and real-time system troubleshooting under high-pressure environments.

OTHER INTERESTS

- **Music:** Piano, Saxophone, Drums, Bass Guitar
- **Creative Technology:** Music Production, Sound Design, Video Editing
- **Sports:** Football, Basketball